Burden-Free Distillation From Foundation Model for Efficient Remote Sensing Change Detection

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***Abstract*— Applying vision foundation models to remote sensing change detection (CD) has attracted extensive research attention. These studies employ inherent general knowledge from vision foundation models to enhance CD performance. Existing methods explicitly employ the foundation model as a feature extractor while designing additional learnable modules to bridge the task gap. However, these methods substantially increase the computational burden and memory demand in the inference. This article therefore focuses on addressing the core challenge of effectively leveraging the knowledge from vision foundation models to enhance CD performance while maintaining computational efficiency. Instead of explicitly utilizing the foundation model, we propose burden-free distillation (BFD), an architecture-agnostic foundation model-based distillation framework for efficient CD. BFD transfers the general knowledge from foundation models to task-specific models, thereby eliminating the dependency on foundation models during inference. Specifically, BFD transfers the foundation model knowledge through dual-temporal feature matching (DFM) module. This module enables multilevel feature alignment by computing pixel-wise spatial similarity between the foundation models' general features and the CD models' task-specific features. Additionally, we leverage patch contrastive distillation (PCD), which transfers localized structural patterns to the CD model to further mitigate task discrepancies between foundation models and CD models. We conduct extensive experiments across multiple foundation models and CD architectures; experimental results demonstrate that BFD effectively adapts the knowledge of foundation models to CD tasks without additional computational burden. Compared to other foundation model-based CD methods, BFD reduces the model parameters by 80.3% and improves IoU by 1.78% on the S2Looking dataset. The code is available at https://github.com/Younger-hua/Burden-Free-Distillation**

***Index Terms*— Change detection (CD), contrastive distillation, foundation models, knowledge distillation (KD).**

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I. Introduction

CHANGE detection (CD) is an essential and widely used task in remote sensing, finding broad applications in urban planning [1], [2], natural disaster assessment [3], agricultural surveys, and land cover monitoring [4]. The task aims to detect and analyze temporal changes occurring over time in the same geographical area [5]. The core principle of CD implementation lies in identifying semantic changes for specific categories of interest while filtering out irrelevant semantic changes [6]. However, achieving accurate CD is extremely challenging due to multiple factors, including differences in image resolution, noise interference, registration errors, and illumination variations between images [7].

Facing these challenges, traditional CD methods including image difference [8], change vector analysis [9], and principal component analysis [10] calculate pixel or feature differences to identify changed areas. While relying on handcrafted features, these methods struggle to accurately detect complex scenes [11], [12]. In recent years, learning-based methods have significantly advanced the development of CD. Cheng et al. [13] adopt Siamese architectures combined attention mechanisms [14], [15] with convolutional neural network (CNN) to propose CNN-based CD models, which have greatly improved the accuracy and efficiency of CD. Ma et al. [16] and Lei et al. [17] utilize vision transformers (ViT) as encoders to aggregate features and capture multitemporal semantic changes, integrating multiregion attention mechanisms and boundary optimization for high-precision detection. Despite the dramatic increase in model parameters, CD models have not achieved proportional performance improvements [18], [19]. Moreover, while learning-based methods generally require large amounts of training data, the pixel-level annotation process for CD remains

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particularly challenging. This limited data availability consequently restricts the performance enhancement of CD models [20].

Foundation models, pretrained on extensive datasets, demonstrate superior generalization capabilities in downstream tasks compared to small-scale models designed for specific tasks or domains [21]. These foundation models can effectively learn generalizable feature representations, reducing reliance on task-specific labeled data. However, given limited data and computational resources, training domain-specific foundation models from scratch to achieve comparable performance remains prohibitively challenging [22].

Recent research explores the parameter-efficient fine-tuning (PEFT) [23] paradigm [24], which explicitly leverages foundation models while integrating additional modules to enhance downstream task performance. As illustrated in Fig. 1, PEFT methods primarily employ adapter and side networks for remote sensing CD. Adapter-based methods incorporate additional learnable modules into frozen foundation models to enable lightweight fine-tuning for CD, whereas side network-based methods utilize bridge modules to align foundation model features and inject foundation model features into the side CD model. Although these methods enhance remote sensing CD performance, they rely on the foundation models as a feature extractor and integrate additional learnable modules for inference, which leads to high computational and memory costs beyond the already resource-intensive foundation model. Such limitations hinder their practical deployment on resource-constrained devices. To address this issue, our work mainly explores how to effectively leverage vision foundation model knowledge to boost CD performance while maintaining efficiency. As shown in Fig. 2, the foundation model serves as the teacher and the CD model serves as the student. We introduce knowledge distillation (KD) to facilitate knowledge transfer from foundation models, avoiding explicitly using foundation models in the inference.

With the above key designs, we propose burden-free distillation (BFD) method for remote sensing CD. This framework effectively transfers task-related knowledge from the foundation model to the CD model through a dual-temporal feature matching (DFM). The DFM module achieves multilevel feature alignment by computing pixel-wise spatial similarity between general features (from the foundation model) and task-specific features (from the CD model), bridging the gap between the two models. Furthermore, while foundation models are good at capturing global semantic information, they often lack explicit modeling of local spatial structures, which can degrade performance in fine-grained CD tasks. To solve this problem, we introduce patch contrastive distillation (PCD), a mechanism designed to transfer structured local context information from the foundation model to the CD model, enabling precise semantic change localization.

The main contributions of this article are as follows.

1) This article proposes a novel BFD method for remote sensing CD. Unlike existing approaches, BFD transfers task-agnostic feature knowledge from foundation models to lightweight CD models while eliminating dependency on foundation models during inference, thereby avoiding substantial computational overhead.
2) This article proposes DFM module that aligns foundation model features with CD model features by computing pixel-wise spatial similarity, thereby distilling task-relevant knowledge from the foundation model into the CD model. Furthermore, to address the limited local structural awareness of foundation models, we introduce a PCD loss, which transfers structured local contextual information from the teacher (foundation model) to the student (CD model), significantly enhancing the CD model's ability to capture fine-grained semantic changes.
3) Extensive experimental results demonstrate the effectiveness of our method without introducing additional inference burden.

The remainder of this article is organized as follows. Section II reviews related work on CD, KD, foundation model, and tuning. Section III introduces the fundamentals of CD and feature distillation. Section IV details the implementation of our proposed method. Section V presents experimental results and comprehensive analysis. Section VI concludes this article.

II. Related Work

A. Change Detection

In recent years, CNNs have been widely used for CD due to their ability to capture contextual information. Initially, Daudt et al. [25] employed UNet-based Siamese CNN encoders to extract multitemporal features and embed them into change representation space. Chen and Shi [1] proposed a Siamese-based spatial-temporal attention network that combines metric learning with self-attention mechanisms to enhance the modeling of spatial-temporal relationships. Fang et al. [26] designed a densely connected Siamese network for CD, which preserves high-resolution fine-grained features through dense skip connections between the encoder and decoder. Recent works have also emphasized the importance of edge information in improving CD accuracy. For example, Bai et al. [27] introduced prior knowledge of edge structures to refine change maps. Lei et al. [17] incorporated boundary extraction constraints into a Siamese network to strengthen feature discriminability.

Recently, Transformer [28] has become popular in the RS community due to its ability to build long-range dependencies. For instance, BiT [29] employed ResNet as a feature extractor and integrated the transformer-based decoder to refine the change maps. ChangeFormer (CF) [18], and EATDer [16] replaced the CNN backbone entirely with ViT to capture temporal dependencies, demonstrating superior performance. On the other hand, some studies have proposed dual-branch structures combining CNNs and Transformers. Wnet [30] adopts parallel Siamese CNNs and Transformers for feature fusion. However, CTD-Former [31] facilitated cross-temporal feature interaction by exchanging the statistical information of bitemporal features. These methods have stacked more modules to achieve a slight improvement. Despite the increasing model sizes in CD, performance improvements are very

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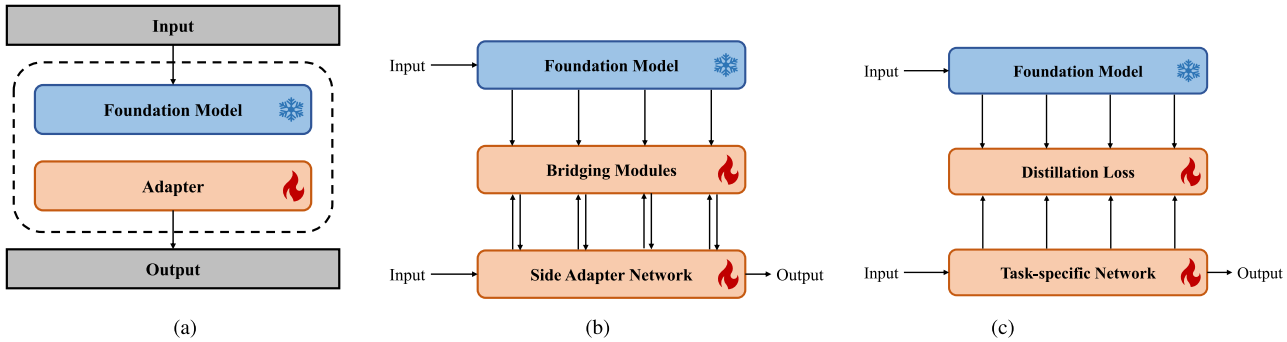


Fig. 1. Comparison of our BFD framework and existing PEFT paradigms for remote sensing CD. (a) Adapter-based methods introduce additional learnable modules to integrate with frozen foundation model for lightweight fine-tuning on CD task. (b) Side network-based methods employ bridge modules to align foundation model features and inject them into the CD model. (c) BFD eliminates the dependency on foundation models during inference by designing distillation loss to transfer general feature knowledge from the foundation model to the task-specific CD model without introducing computational burden.

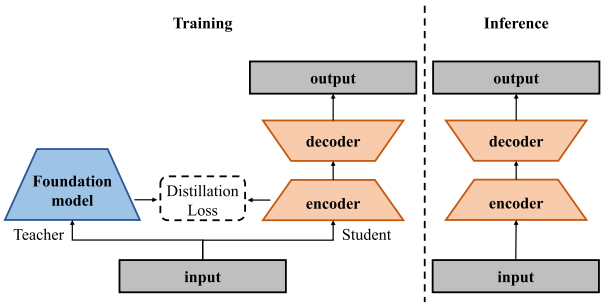


Fig. 2. Overview of our proposed BFD for remote sensing CD. During training, BFD aligns features between the foundation model and the CD model via a distillation loss to transfer task-agnostic knowledge. During inference, only the lightweight CD model is deployed, eliminating computational dependence on the foundation model.

limited. In this article, we propose a foundation model-based KD framework for efficient remote sensing CD.

B. Foundation Model and Tuning

Deep learning models, particularly for dense prediction tasks, typically require extensive labeled training data. To mitigate this dependency, researchers have explored pretrained foundation models on task-agnostic datasets which enable generalization to downstream tasks with minimal task-specific training. A notable example is CLIP [21], a vision–language pretrained method that trains dual encoders to maximize similarity between visual and textual embeddings of matched image–text pairs while minimizing similarity for mismatched pairs. CLIP demonstrates remarkable zero-shot generalization capabilities, achieving classification performance comparable to supervised CNNs without task-specific fine-tuning. Beyond vision–language models, SAM [32] advances segmentation by training on an extensive dataset (11 million images, 1 billion masks) to produce a universal segmentation model.

Liu et al. [33] proposed RemoteCLIP, the first vision–language foundation model tailored for remote sensing. However, RemoteCLIP's improvement on some remote sensing tasks is limited due to the difficulty of constructing remote sensing image–text pairs. Unlike natural images, where billions of training samples are readily available, remote sensing data scarcity makes it difficult to train foundation models from scratch. Facing this challenge, Houlsby et al. [23] have explored PEFT paradigms to adapt existing foundation models to specific downstream tasks. PEFT freezes most pretrained parameters while introducing lightweight learnable components (e.g., adapters) for partial fine-tuning. Li et al. [20] proposed BAN, which injects general features from foundation models into a side adapter network. SAM-CD [34] uses a frozen Fast-SAM [35] encoder and fine-tunes only the prediction head. While these methods achieve impressive performance, they inherently require integrating foundation models during inference, significantly increasing computational costs and limiting their practicality in resource-constrained scenarios. In contrast, our work leverages KD to transfer the rich knowledge of foundation models to lightweight CD models, enhancing performance without introducing additional parameters.

C. Knowledge Distillation

KD, first introduced by Hinton et al. [36], enables the transfer of knowledge from a high-capacity teacher model to a compact student model. Several works proposed feature-based distillation methods to achieve knowledge transfer by mimicking the teacher's features [37], [38], [39]. For instance, FitNet [37] unified the feature dimension of the student and teacher by mapping operators and then performed the knowledge transfer in intermediate features. SemCKD [38] proposed an attention-based soft layer association mechanism for cross-layer distillation to deal with the negative regularization caused by semantic mismatch between student and teacher networks. KR [39] proposed the feature transformation by cross-stage connection paths between teacher and student networks for KD. However, most of above methods are designed for image classification and unsuitable for pixel-level tasks such as semantic segmentation and remote sensing CD.

Recent KD methods often consider context information and class relations as knowledge. CWD [40] proposed channel-wise distillation to guide the student mimicking the teacher's semantics along the channel dimension. SKD [41] directly measured the pixel-level similarity between the teacher and the student. IFVD [42] measured the distance between different class prototypes and forced the student

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to imitate the teacher's intraclass relations. However, conventional KD paradigms are primarily designed for model lightweighting within the same data or task and cannot directly transfer knowledge from foundation models to specific downstream tasks. To bridge this gap, we propose a DFM module and a task-aware distillation framework that aligns cross-task representations between foundation models and CD models, effectively addressing their inherent task disparities.

III. Preliminaries

A. Change Detection

Deep learning-based CD methods utilize a Siamese encoder to extract features from different temporal phases and then embed them into the feature representation space [43]. Let (x_1, x_2) denote the temporal images, and $E(\cdot)$ and $D(\cdot)$ denote the encoder and decoder networks. The predicted change map m can be expressed as

$$\begin{aligned} F_1^s &= E(x_1) \\ F_2^s &= E(x_2) \\ m &= D\left(F_1^s, F_2^s\right). \end{aligned} \tag{1}$$

B. Feature Distillation

We apply feature distillation to transfer the general feature knowledge from the foundation model. The original features-based distillation loss is defined as

$$\mathcal{L}_{\text{feat}} = \left\| f^t - \phi\left(f^s\right) \right\|_2 \tag{2}$$

where f^t and f^s denote the teacher (foundation model) and student (CD model) features, respectively. ϕ represents the projection layer that aligns the dimension of the feature of the student and teacher networks. Previous work [37], [39] minimizes the feature distance between teacher and student trained with the same tasks, which fail to achieve optimal knowledge transfer from teacher–student structures training on different tasks. This article focuses on transferring the general knowledge from pretrained foundation models to the remote sensing domain to achieve more accurate CD predictions.

IV. Methodology

The framework of our proposed BFD for remote sensing CD is illustrated in Fig. 3. In the training process, we first extract multilevel features from the foundation model and CD model, respectively, and perform their feature semantic similarity through DFM module. Then, we reweight the CD features for matching based on semantic similarity. We use distillation loss L_{BFD} to align the matched features with the general features. The network structure and technical details of our method are described as follows.

A. General Feature Extraction

CLIP is pretrained on 400 million image–text pairs to achieve robust visual feature extraction beyond domain-specific biases. In remote sensing CD tasks, since bitemporal images need to be received as inputs, we employ CLIP (ViT-B/16) [21], [44] with shared parameters (i.e., a Siamese network) to extract features simultaneously. The existing ViT CLIP models are designed for low-resolution images (e.g., 224 × 224 or 336 × 336). Self-attention mechanisms in ViT are sensitive to changes in resolution, directly inputting high-resolution remote sensing images into the foundation model will disrupt the prelearned patch relationships and position embeddings, leading to performance degradation. To address the input resolution mismatch, we adopt the asymmetric resolution input strategy (ARIS) following with [20] and [45], we downsample the input images to the pretrained resolution in the foundation models and keep original resolution in the task-specific models. This process is formulated as

$$\begin{aligned} F_1^t &= \text{FM}(\text{ARIS}(x_1)) \\ F_2^t &= \text{FM}(\text{ARIS}(x_2)) \end{aligned} \tag{3}$$

where F_1^t and F_2^t represent the general features of the two temporal images, respectively.

B. Dual-Temporal Feature Matching

To tackle data distribution shift, feature resolution difference, and task gap, we adopt feature normalization (FN) and propose a semantic similarity mechanism to match task-relevant general features with task-specific features, as shown in Fig. 3.

For the obtained bitemporal general features $F_i^t \in \mathbb{R}^{H_t W_t \times C_t}$ and task-specific features $F_i^s \in \mathbb{R}^{H_s W_s \times C_s}$, where i represents ith temporal, $i \in \{1, 2\}$. To eliminate feature shift between the foundation model and the CD model caused by data distribution discrepancies (e.g., illumination, scale, and style variations), we standardize the general features through FN. It can be represented as follows:

$$\tilde{F}_i^t = \phi\left(\left(F_i^t - \mu\left(F_i^t\right)\right)/\sigma\left(F_i^t\right)\right) \tag{4}$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ compute the mean and variance, respectively, and ϕ is a 1 × 1 convolutional layer for channel dimension alignment between foundation model features F_i^t and CD features F_i^s.

CD usually requires capturing pixel-level differences between objects of interest in bitemporal images. However, the objective of the foundation model learning pursues high-level semantic understanding (e.g., object classification, scene recognition), and the learned features tend to emphasize global semantics. To bridge this gap, BFD implements an attention mechanism that enables the CD model to learn robust task-relevant feature representations based on semantic similarity, thereby bridging the gap between CD and pretraining tasks. Specifically, BFD employs a dot-product operation to obtain the semantic similarity S_i, which captures the associations between the general and CD features

$$S_i = \tilde{F}_i^t \times F_i^s \qquad S_i \in \mathbb{R}^{H_t W_t \times H_s W_s}. \tag{5}$$

Then, we apply a softmax function to match the features of each pixel

$$S_i = \text{Softmax}\left(\tilde{F}_i^t \times F_i^s\right). \tag{6}$$

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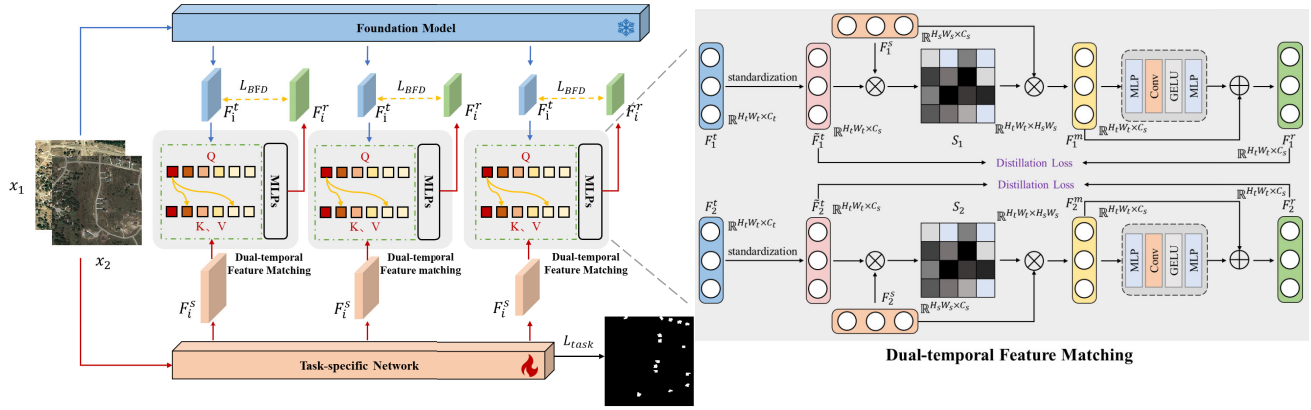


Fig. 3. Architecture of BFD for remote sensing CD. First, features are extracted independently of both the foundation model and the CD model. The DFM module then computes a semantic similarity map between the foundation model's general features and the CD model's task-specific features. Leveraging this similarity map, DFM aligns the task-specific features with the foundation model's general features by projecting them into a shared representation space, thereby bridging the task gap and enabling efficient knowledge transfer.

Leveraging the semantic similarity map S_i, we aggregate these relevant semantics to adaptively match the CD features

$$F_i^m = S_i \times F_i^s \qquad F_i^m \in \mathbb{R}^{H_t W_t \times C_s}. \tag{7}$$

Precise spatial location information is crucial for accurately delineating change boundaries, such as building edges and road extensions. While F^m contains task-relevant general feature representations, it loses the original high-resolution feature representations, resulting in the loss and misalignment of spatial information. To address this, we introduce 3×3 convolution and multilayer perceptron (MLP) to enhance the spatial awareness of the features

$$F_i^r = \mathrm{MLP}\big(\mathrm{GELU}\big(\mathrm{Conv}_{3\times3}\big(\mathrm{MLP}\big(F_i^m\big)\big)\big)\big) + F_i^m \tag{8}$$

where MLP denotes the linear layer.

C. Distillation Loss

We aim to align F_i^r and $\tilde{F}_i^t$ to ensure that the task-relevant feature pixels achieve robust representations comparable to those of the foundation model. In the training process, we minimize task loss and distillation loss to optimize the network parameters.

For distillation loss, we first utilize the traditional pixel-wise feature alignment and sum the feature distances of the two temporal with identical weights

$$\mathcal{L}_{\text{feat}} = \sum_{i=1}^{2} \left\| F_i^r - \tilde{F}_i^t \right\|_2. \tag{9}$$

The traditional feature alignment loss $\mathcal{L}_{\text{feat}}$ can only capture task-beneficial global semantic information, making it difficult to retain local detail information that is crucial for CD. To address this issue, we define a patch distillation contrastive loss $\mathcal{L}_{\text{patch}}$ to transfer the structural feature knowledge of local regions from the foundation model, as shown in Fig. 4.

First, we define positive pairs as $F_i^r(u)$ and $\tilde{F}_i^t(v)$ which have the same index $u = v$. Negative pairs can be defined as: $F_i^r(u)$ and $\tilde{F}_i^t(v)$ with different indexes $u \neq v$, where u and v are location indices in spatial. Based on these definitions, the general contrastive loss [46] is formulated as

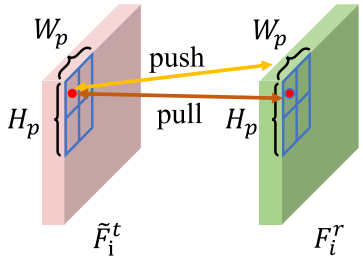


Fig. 4. Detail of PCD. We first partition both feature maps into nonoverlapping (H_p, W_p) local patches. In each patch, contrastive learning is performed to align local semantics. The red arrows indicate constructing positive pairs with same indexes, while the yellow arrows indicate constructing negative pairs with different indexes.

$$l_{\text{contrast}} = -\log \frac{\exp\big(-d\big(F_i^r(u), \tilde{F}_i^t(u)\big)/\tau\big)}{\sum_{v=1}^{N} \mathbb{1}_{u\neq v} \exp\big(-d\big(F_i^r(u), \tilde{F}_i^t(v)\big)/\tau\big)} \tag{10}$$

where τ is the temperature, N is the number of contrastive pairs, and $\mathbb{1}_{u\neq v} \in \{0, 1\}$ is an indicator function which equals 1 iff $u \neq v$. $d(\cdot, \cdot)$ is Euclidean distance. However, using every pixel of the features as a contrastive pair incurs significant computational costs and fails to capture the correlations between local neighborhoods. We divide the feature maps into patches and perform the contrastive operation for each patch [47]. Then, patch distillation contrastive loss can be formulated as

$$\mathcal{L}_{\text{patch}} = \sum_{i=1}^{2} \sum_{p=1}^{M} \sum_{j=1}^{H_p} \sum_{k=1}^{W_p} l_{\text{contrast}}\big(F_i^r(p, j, k), \tilde{F}_i^t(p)\big) \tag{11}$$

where H_p and W_p denote the size of the feature patch. $M = (\mathrm{HW}/H_p W_p)$. We combine feature alignment and patch contrastive losses as distillation loss

$$\mathcal{L}_{\text{BFD}} = \alpha \mathcal{L}_{\text{feat}} + \beta \mathcal{L}_{\text{patch}} \tag{12}$$

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Algorithm 1 Training BFD on One Mini-Batch

Input: Bitemporal image pairs $\{x_1, x_2\}$, initial the pretrained foundation model FM, the CD encoder $E(\cdot)$, the CD decoder $D(\cdot)$, and DFM.

Output: The prediction change map Y.

1: **Step 1:** Input image pairs (x_1, x_2) into FM with ARIS, and obtain feature F_1^t and F_2^t, according to Eq.3.
2: **Step 2:** Input image pairs (x_1, x_2) into $E(\cdot)$, and obtain feature F_1^s and F_2^s, according to Eq.1.
3: **Step 3:** Input $\{F_1^t, F_2^t, F_1^s, F_2^s\}$ into DFM and get feature $\{F_1^r, F_2^r, \tilde{F}_1^t, \tilde{F}_2^t\}$, according to Eq.4, Eq.7 and Eq.8.
4: **Step 4:** According to $\{F_1^s, F_2^s\}$ and $D(\cdot)$, and obtain change map Y.
5: **Step 5:** Compute the training loss, according to Eq. 14.
6: **Step 6:** Loss backpropagation and update the network parameters.

where α and β are feature spatial and PCD loss weights. For task loss, it is defined as

$$\mathcal{L}_{\text{task}} = -\frac{1}{H \times W} \sum_{h=1,w=1}^{H,W} Y(h,w)\log(P(h,w)) \tag{13}$$

where H and W denote the height and width of the output. $Y(h,w)$ and $P(h,w)$ represent the label and prediction for the pixel at (h,w), respectively. Finally, the overall training loss can be defined as

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \mathcal{L}_{\text{BFD}}. \tag{14}$$

The details of BFD training process on mini-batch is shown in Algorithm 1.

V. Experiments

In this section, we evaluate the effectiveness of the proposed method on multiple benchmark datasets. We adopt two CD architectures as student models: BiT [29] and CF [18]. We compare the complexity of various CD models and the performance of BFD trained with different foundation models. The dataset introduction, implementation details, and experimental results are described below.

A. Datasets

We conduct experiments on four standard CD datasets: the LEVIR-CD [1] dataset, WHU-CD [2] dataset, CLCD dataset, and S2Looking [50] dataset. Additionally, we utilize only the weights of the foundation model for distillation without using any extra data. We briefly introduce these datasets as below.

1) LEVIR-CD: It contains 637 pairs of bitemporal remote sensing images from Google Earth, with 31 333 annotated change instances. Each image pair has dimensions of 1024 × 1024 pixels with a spatial resolution of 0.5 m. The training, validation, and test sets have 445, 64, and 128 image pairs, respectively.

2) WHU-CD: It consists of aerial images collected in Christchurch, New Zealand between 2012 and 2016. The original images have 32507 × 15354 pixels and a resolution of 0.2 m. We first crop the image patches of 256 × 256 pixels. Following [51], we divide the dataset into 5947/743/744 samples for training, validation, and test sets, respectively.

3) CLCD: It focuses on cropland changes and comprises 600 pairs of Gaofen-2 satellite images collected in Guangdong Province, China. Each image has 512 × 512 pixels with a spatial resolution ranging from 0.5 to 2 m. The images are divided into 320, 120, and 120 pairs for the training, validation, and test sets, respectively.

4) S2Looking: It is a building CD benchmark dataset covering rural areas worldwide. It consists of 5000 pairs of images, each with 1024 × 1024 pixels and a spatial resolution ranging from 0.5 to 0.8 m. Unlike other CD datasets, many image pairs in the S2Looking dataset are captured from different imaging angles.

B. Implementation Details

During the training process, we compute the feature distillation loss for each backbone layer and use the AdamW optimizer. The learning rate is 2×10^{-4}. The poly schedule is employed to reduce the learning rate at each iteration. Data augmentation techniques include random cropping (crop size is set to 512 × 512 for the LEVIR-CD and S2Looking datasets), flipping, and photometric distortion. We train for 40, 80, and 40 K iterations on the LEVIR-CD, S2Looking, and CLCD datasets, respectively, and 100 epochs on the WHU-CD dataset. All experiments are conducted on an NVIDIA GeForce RTX 3090 GPU with the batch size set to 8. By default, α, β, and τ are set to 2, 0.1, and 0.07, respectively. Patch sizes (H_p, W_p) set to (2, 2). All models are trained with OpenCD [52].

C. Evaluation Metrics

We adopt the intersection over union (IoU), F_1 score, precision (P), and recall (R) as evaluation metrics, which are calculated as follows:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

$$F_1 = 2 \times \frac{P \times R}{P + R} \tag{17}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{18}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{19}$$

where TP, FP, TN, and FN indicate true positive, false positive, true negative, and false negative, respectively.

D. Comparison Experiments

1) Quantitative Results: We conduct extensive comparisons between BFD and other CD methods, and the comparison methods include **CNN-based** approaches, including

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FC-Siam [25], and SNUNet [26], as well as **Transformer-based** models, including BIT [29], CF [18], CTD-Former [31], CGNet [49], and EATDer [16]. Additionally, we also compare with **Foundation model-based** methods, SAM-CD [34], and BAN [20].

As shown in Table I, BFD demonstrates consistent performance improvements across architectural paradigms. For the LEVIR-CD dataset, BFD-BIT achieves up to 0.95% improvement in mIoU, and BFD-CF achieves up to 0.82% improvement in mIoU. After distilling general knowledge, the performance of BFD-CF on LEVIR-CD surpasses the state-of-the-art (SOTA) method, SAM-CD. On the WHU-CD dataset, BFD-BIT achieves 95.89% in mF1 and 92.35% in mIoU, and BFD-CF achieves 97.23% in mF1 and 94.72% in mIoU. Notably, existing methods improve performance by stacking more modules, which increases the memory and computational costs, yet yields minimal gains. BFD introduces the general knowledge of foundation models from a new perspective to improve the performance of existing CD methods without additional inference burden.

To validate the generalization and robustness of our proposed method, we conduct additional evaluations on the challenging CLCD and S2Looking datasets containing complex illumination variations. BFD-BIT achieves improvements of 3.43% and 0.87% in mIoU, while BFD-CF achieves improvements of up to 1.57% and 2.39% in mIoU, as reported in Tables II and III. BFD significantly improves the Recall of BIT and CF, demonstrating that BFD effectively extracts task-beneficial knowledge from the foundation model and enhances the discriminability of semantic feature changes.

2) Semi-Supervised CD: To validate the transfer ability of robust feature knowledge from the foundation model to CD task, we conduct experiments in the semi-supervised CD setting. Semi-supervised CD aims to train the CD model with a small amount of labeled data and a large amount of unlabeled data. The evaluation metric is the IoU of change areas. In this experiment, we train only partial labels without utilizing the unlabeled data. Table IV compares the accuracy of different CD methods trained with a limited proportion of training data on the WHU-CD dataset. Besides the baseline method "Sup. only," we also compare with semi-supervised methods, where AdvNet [53] and s4GAN [54] are semi-supervised semantic segmentation methods. SemiCDNet [55] and SemiCD [51] are SOTA semi-supervised CD methods. Apart from SemiCD, BFD outperforms other semi-supervised methods with 5% and 10% of the labeled data. Due to the limited data, BFD cannot extract sufficient relevant knowledge from the remote sensing domain, resulting in its performance being inferior to SemiCD. BFD achieves 77.24% IoU with 20% of the labeled data, which is 2.44% superior to SemiCD. And with 40% of the labeled data, BFD achieves 81.47% IoU, surpassing SemiCD by 4.27%. These results demonstrate that BFD transfers extensive knowledge of the foundation model, effectively reducing the dependence of remote sensing CD on data labels, which is beneficial to the application of remote sensing tasks.

3) Visualization Results: In Fig. 5, we present several examples of the results obtained on the LEVIR-CD dataset. We use red and yellow to indicate missed and false detections, respectively. As shown in the fourth and sixth rows of the figure, BFD demonstrates superior performance in detecting changes within buildings exhibiting unique appearances and has a stronger ability to capture changes, making the features of the changed areas easier to identify. Additionally, Fig. 6 illustrates more complex CD scenarios on the CLCD dataset. BFD demonstrates fewer false positives when dealing with intricate changes. These observations suggest that by leveraging knowledge from the foundation model, BFD achieves improved discrimination between temporal variations and semantic changes.

4) Computational Costs: We compare the model parameters and performance of BFD with other methods, as illustrated in Fig. 7. The analysis reveals that existing approaches tend to pursue increasingly complex models to improve CD performance, but the gains are minimal while the model parameters grow exponentially. For instance, SAM-CD achieves 91.68% mIoU on LEVIR-CD with 70.59M parameters, while BFD-CF surpasses this performance (92.13% mIoU) using only 13.94M parameters, reducing the parameter count by 80.3%. Similarly, BFD-BIT maintains the original parameter size of 2.99M while achieving 91.78% mIoU, outperforming heavier models like CGNet (33.68M) and EATDer (6.61M). These results demonstrate that BFD effectively adapts foundation models to remote sensing CD tasks, achieving the optimal trade-off between efficiency and performance.

E. Ablation Study

In this section, we first evaluate the performance of BFD under varying input resolutions of the foundation model. Subsequently, we compare the proposed DFM module with existing feature alignment methods. We further analyze the sensitivity of the performance of BFD with different hyperparameter configurations, particularly the loss weights α and β in (12). Additionally, we examine the effects of alternative foundation model architectures and varying distillation layer selections on model performance. All experiments are conducted on the Levir-CD dataset, using CLIP ViT-B/16 as the foundation model and BIT as the CD model. Unless explicitly stated, all other experimental parameters remain consistent with the implementation specifications outlined in Section V-B.

1) Effects of ARIS: We perform ablation studies on ViT-based foundation models with varying input resolutions to validate the ARIS mechanism. Specifically, We employ two configurations: CLIP ViT-L/14 pretrained at 336 × 336 and CLIP ViT-B/16 pretrained as 224 × 224 resolution. As shown in Table V, the performance is optimal when the input resolution of the foundation model matches the resolution used during the pretrained phase, while performance degrades at other resolutions. We also unify the input resolution of the foundation model and the CD model to 512 × 512, but this high-resolution setting still yields inferior performance compared to the original pretrained resolution. This evidence highlights that maintaining resolution consistency between inference inputs and pretraining configurations critically affects model efficacy. Changes in resolution disrupt the prelearned patch relationships and positional embeddings,

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TABLE I

COMPARISON RESULTS WITH OTHER CD METHODS ON PARAM (MODEL PARAMETERS(M)), P (%), R (%), mF$_1$(%) AND mIoU(%) ON THE LEVIR-CD AND WHU-CD DATASETS. * MEANS THE REIMPLEMENTED RESULTS WITH [48]

Method	Param	Levir-CD				WHU-CD			
		P	R	mF_1	$mIoU$	P	R	mF_1	$mIoU$
FC-Siam-diff [25]	1.66	92.93	87.02	89.73	82.67	94.52	89.32	91.74	85.65
FC-Siam-conc [25]	2.74	92.03	89.82	90.89	84.34	93.94	92.83	93.37	88.19
SNUNet [26]	1.35	93.83	90.11	91.88	85.84	92.61	83.26	87.30	79.40
CTD-Former [31]	3.85	93.60	91.85	92.71	87.11	96.74	97.03	96.89	94.11
CGNet [49]	33.68	95.95	94.95	95.44	91.58	98.04	96.14	97.07	94.44
EATDer [16]	6.61	96.29	92.70	94.41	89.84	95.57	93.06	94.28	89.64
SAM-CD [34]	70.59	95.87	**95.14**	95.50	91.68	97.97	**97.20**	**97.58**	**95.36**
BIT* [29]	2.99	96.13	93.93	95.00	90.83	93.00	95.93	94.88	90.64
BFD-BIT	2.99	96.13	95.00	95.56	91.78 ↑0.95	96.02	95.75	95.89	92.35 ↑1.71
CF* [18]	13.94	96.32	94.30	95.28	91.31	97.9	95.59	96.71	93.8
BAN-CF-b0 [20]	68.88	**96.87**	93.10	94.89	90.65	98.10	95.47	96.75	93.86
BFD-CF-b1	13.94	96.51	95.05	**95.77**	**92.13** ↑0.82	**98.15**	96.34	97.23	94.72 ↑0.92

TABLE II

COMPARISON RESULTS OF DIFFERENT OTHER CD METHODS ON P(%), R(%), mF$_1$(%), AND mIoU(%) ON THE CLCD DATASET. * MEANS THE REIMPLEMENTED RESULTS WITH [48]

Method	P	R	OA	mF_1	$mIoU$
FC-Siam-diff [25]	83.13	74.60	94.73	78.10	68.19
FC-Siam-conc [25]	81.65	79.94	94.84	80.77	70.98
SNUNet [26]	85.92	82.62	95.84	84.19	75.11
CTD-Former [31]	87.29	83.17	96.12	85.08	76.24
CGNet [49]	85.87	85.30	96.00	85.58	76.83
EATDer [16]	87.16	77.18	93.84	81.16	71.18
SAM-CD [34]	**88.25**	85.65	96.26	86.89	78.53
BIT* [29]	75.30	79.85	93.19	77.34	67.04
BFD-BIT	81.55	79.16	94.78	80.29	70.47 ↑3.43
CF* [18]	86.21	85.78	96.16	85.99	77.37
BFD-CF-b1	87.94	**86.43**	**96.53**	**87.16**	**78.94** ↑1.57

TABLE III

COMPARISON RESULTS OF DIFFERENT OTHER CD METHODS ON P(%), R^c(%), F_1(%), AND IoU(%) ON THE S2LOOKING DATASET. * MEANS THE REIMPLEMENTED RESULTS WITH [48]

Method	P	R	F_1	IoU
FC-Siam-diff [25]	83.49	32.32	46.60	30.38
FC-Siam-conc [25]	68.27	18.52	13.54	-
SNUNet [26]	45.26	50.60	47.78	31.39
CGNet [49]	70.18	59.38	64.33	47.41
EATDer [16]	65.85	54.74	59.78	42.64
SAM-CD [34]	72.80	58.92	65.13	48.29
BiT* [29]	74.65	55.33	63.55	46.57
BFD-BiT	74.77	56.48	64.35	47.44 ↑0.87
CF* [18]	**77.68**	55.25	64.57	47.68
BAN-CF-b1 [20]	74.63	60.30	66.70	50.04
BFD-CF-b1	73.66	**61.0**	**66.73**	**50.07** ↑2.39

and merely increasing the resolution fails to recover the performance loss induced by resolution mismatch.

2) Effects of DFM: We conduct ablation studies to evaluate the effectiveness of different feature alignment methods, as detailed in Table VI. Since the feature resolutions of the foundation model and the CD model are inconsistent, simple resizing strategies (resize FM/resize CD) performs significantly worse than DFM. Specifically, resize FM achieves only 88.64% mF$_1$ and 81.15% mIoU, while resize CD achieves only 93.10% mF$_1$ and 87.72% mIoU. This performance gap stems from the irreversible information loss caused by fixed interpolation operations during resizing, which distorts spatial relationships in feature maps and leads to misaligned spatial information.

In contrast, DFM applies a semantic similarity mechanism to match task-relevant general features with task-specific features, enabling the CD model to better discriminate between true changes and disturbances. Therefore, DFM achieves 95.56% mF$_1$ and 91.78% mIoU. When the convolutional layers and MLP are removed, the performance drops slightly. The convolutional layers enhance local pattern preservation, while the MLP enables nonlinear feature transformation. These components further enhance the spatial awareness of the features, mitigating the degradation caused by resolution reduction.

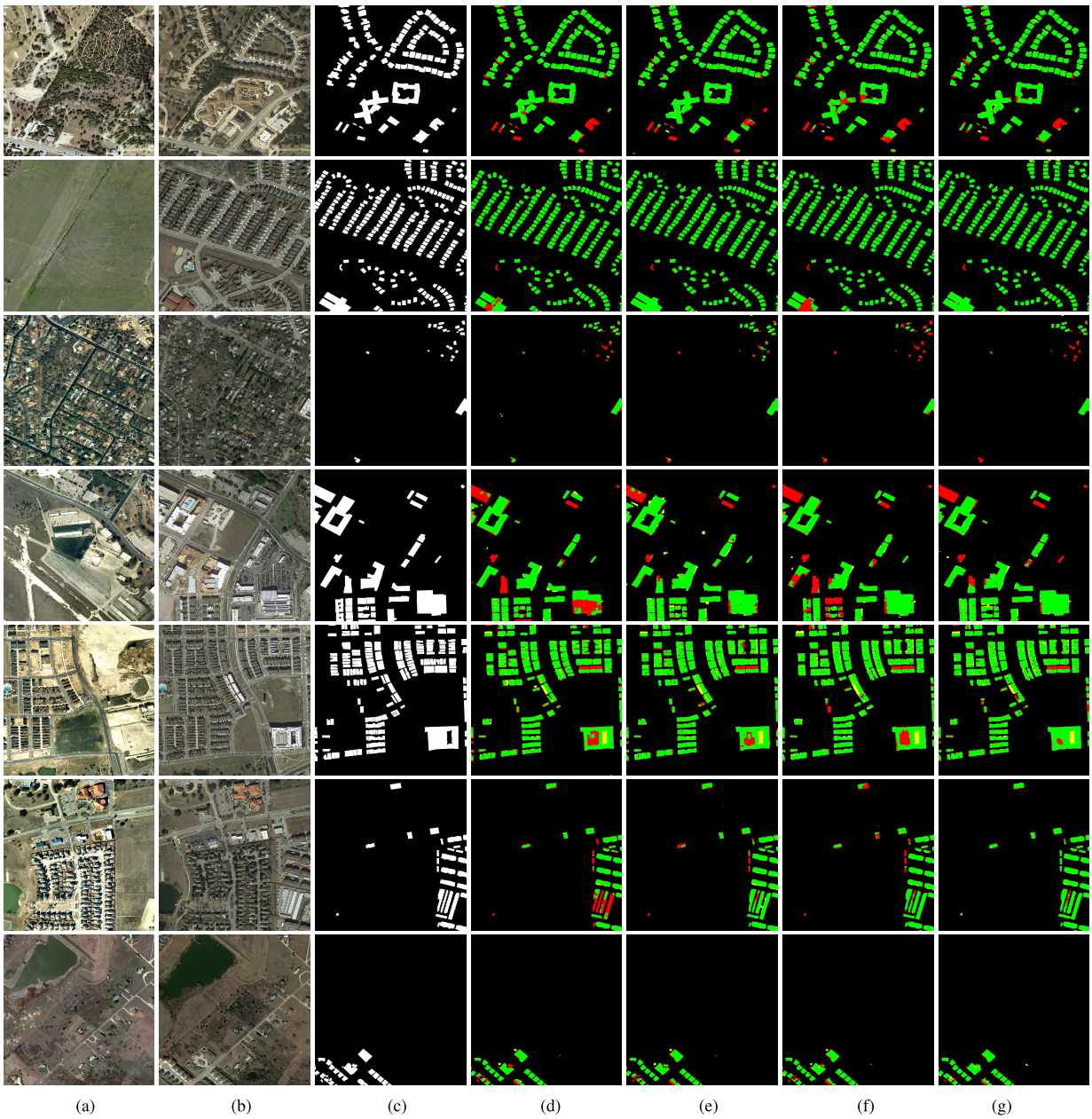
3) Distill Layers: Features from different levels of the foundation models play distinct roles in the distillation process. As summarized in Table VII, we evaluate the effectiveness of distilling features from various layers of the foundation models. In single-layer distillation, the intermediate layer achieves the best performance with 91.76% mIoU, as it balances semantic richness and resolution. Shallow features exhibit limited high-level semantic content, resulting in suboptimal distillation efficacy. Conversely, deeper features suffer from a significant resolution mismatch with the CD model, making it difficult to learn task-specific knowledge effectively. For multilayer distillation, combining features from three levels of the foundation model improves the adaption of CD model to multiscale changes, achieving 91.78% mIoU.

4) Effects of Loss: We perform experiments to show the effects of distillation loss in BFD in Table VIII. $\mathcal{L}_{\text{feat}}$ and $\mathcal{L}_{\text{patch}}$ can significantly improve the performance of BIT, with $\mathcal{L}_{\text{feat}}$ improving by 0.78% and $\mathcal{L}_{\text{patch}}$ improving by 0.71%, and combining them achieves the best performance 91.78% mIoU. In addition, we also conduct ablation experiments on the hyperparameters of the loss. $\mathcal{L}_{\text{feat}}$ primarily governs global feature alignment, ensuring the transfer of task-relevant semantics from the foundation model, thus a larger weight should be applied. $\mathcal{L}_{\text{patch}}$ enhances local pattern preservation and detail refinement. However, when its weight is too large,

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TABLE IV
COMPARISON RESULTS OF SEMI-SUPERVISED CD METHODS ON IoU(%) AND OA(%) ON THE WHU-CD DATASET

Method	5%		10%		20%		40%	
	IoU	*OA*	*IoU*	*OA*	*IoU*	*OA*	*IoU*	*OA*
Sup. only [51]	50.00	97.48	55.70	97.53	65.40	98.20	76.10	98.04
AdvNet [53]	55.10	97.90	61.60	98.11	73.80	98.80	76.60	98.94
s4GAN [54]	18.30	96.69	62.60	98.15	70.80	98.60	76.40	98.96
SemiCDNet [55]	51.70	97.71	62.00	98.16	66.70	98.28	75.90	98.93
SemiCD [51]	**65.80**	**98.37**	**68.10**	**98.47**	74.80	98.84	77.20	98.96
ours	58.64	98.11	64.35	98.42	**77.24**	**99.03**	**81.47**	**99.22**



(a) (b) (c) (d) (e) (f) (g)

Fig. 5. Visualizational results of the existing methods and our proposed BFD on the LEVIR-CD dataset. Green represents true positives, yellow represents false positives, red represents false negatives, and black represents true negatives (a) T_1 image. (b) T_2 image. (c) GT. (d) BIT. (e) BFD-BIT. (f) CF. (g) BFD-CF.

it can cause the model to overly focus on details at the expense of global information. Therefore, it should be applied with a smaller weight. When α is 2, β is 0.1, BFD achieves the best performance.

5) Different Foundation Models: We adapt different foundation models to investigate how various types of general knowledge affect KD. Specifically, we use AugReg [56], fully supervised on ImageNet-21k [58], CLIP [21] trained

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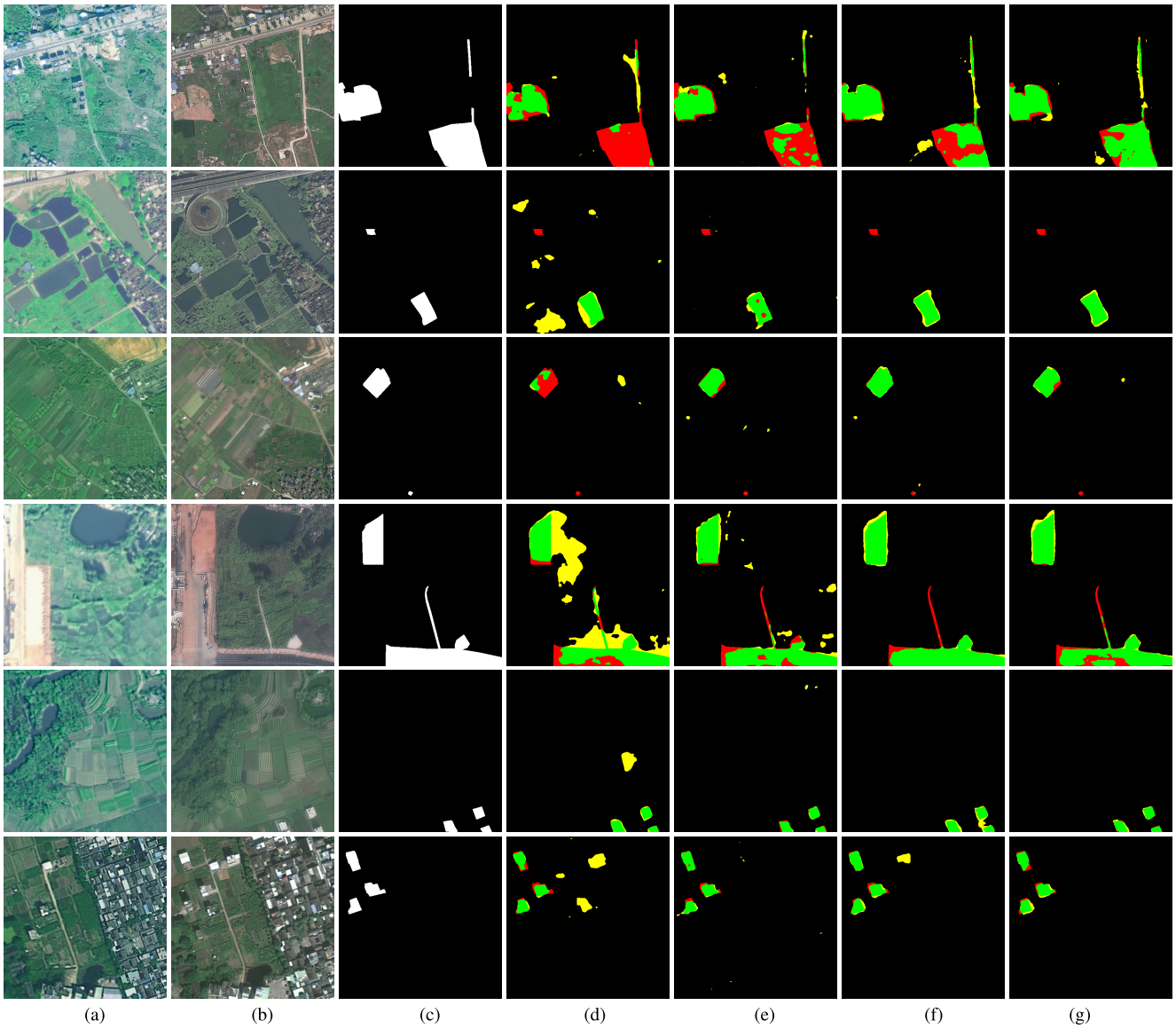


Fig. 6. Visualizational results of the existing methods and our proposed BFD on the CLCD dataset. Green represents true positives, yellow represents false positives, red represents false negatives, and black represents true negatives (a) T_1 image. (b) T_2 image. (c) GT. (d) BIT. (e) BFD-BIT. (f) CF. (g) BFD-CF.

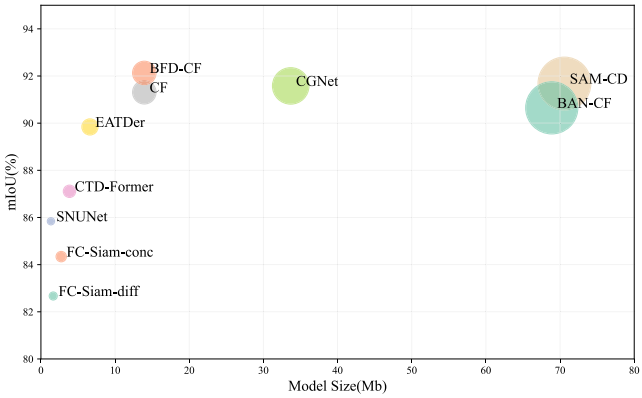


Fig. 7. Comparisons of model parameters and inference performance of BFD with other CD methods on LEVIR-CD dataset.

TABLE V
RESULTS WITH VARIOUS INPUT RESOLUTIONS FOR FOUNDATION MODEL (CLIP VIT-L/14 AND VIT-B/16)

Resolution	ViT-L/14(CLIP) mF1	ViT-L/14(CLIP) mIoU	ViT-B/16(CLIP) mF1	ViT-B/16(CLIP) mIoU
224×224	95.51	91.69	**95.50**	**91.78**
336×336	**95.52**	**91.72**	95.45	91.58
512×512	95.46	91.61	95.35	91.43

TABLE VI
RESULTS WITH DIFFERENT ALIGNMENT METHODS

Alignment	P	R	mF_1	$mIoU$
Resize FM	93.10	85.10	88.64	81.15
Resize CD	94.13	92.12	93.10	87.72
DFM w/o Conv and MLP	96.06	94.83	95.43	91.56
DFM	**96.13**	**95.00**	**95.56**	**91.78**

on the LAION-2B [59] dataset, and RemoteCLIP [33] and GeoRSCLIP [57] trained on remote sensing data to distill various types of knowledge into BIT. As shown in Table IX, the smaller ViT-B architecture generally outperforms the larger ViT-L architecture. We attribute this to the significant

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TABLE VII
RESULTS WITH DIFFERENT DISTILLATION STRATEGIES. CLIP VIT-B/16 IS USED IN ALL EXPERIMENTS

Description	distiller Layers	mF_1	$mIoU$
single-distill	3rd layer	95.40	91.51
	6th layer	95.55	91.76
	9th layer	95.38	91.47
multi-distill	{3,6}-layers	95.45	91.59
	{3,6,9}-layers	**95.56**	**91.78**

TABLE VIII
RESULTS OF LOSS ON VARIOUS HYPERPARAMETERS

Methods	$\mathcal{L}_{feat}$	$\mathcal{L}_{patch}$	α	β	OA	$mIoU$	F_1
BIT(baseline)			-	-	99.06	90.83	95.00
BFD-BIT	✓		2	-	99.13	91.61	95.46
BFD-BIT		✓	-	0.1	99.13	91.54	95.42
BFD-BIT	✓	✓	1	0.1	99.13	91.59	95.45
BFD-BIT	✓	✓	2	0.1	**99.15**	**91.78**	**95.56**
BFD-BIT	✓	✓	5	0.1	99.15	91.73	95.53
BFD-BIT	✓	✓	2	0.5	99.12	91.43	95.35
BFD-BIT	✓	✓	2	1	99.13	91.57	95.44

TABLE IX
RESULTS WITH DIFFERENT FOUNDATION MODELS

Foundation Model	Backbone	P	R	mF_1	$mIoU$
AugReg [56]	ViT-B/16	96.18	94.88	95.52	91.70
AugReg [56]	ViT-L/16	96.13	94.89	95.50	91.68
CLIP [21]	ViT-B/16	96.13	95.00	95.50	**91.78**
CLIP [21]	ViT-L/14	96.02	95.04	95.52	91.72
RemoteCLIP [33]	ViT-B/32	96.07	**95.07**	**95.56**	**91.78**
RemoteCLIP [33]	ViT-L/14	**96.21**	94.73	95.46	91.61
GeoRSCLIP [57]	ViT-B/32	96.16	94.87	95.51	91.68
GeoRSCLIP [57]	ViT-L/14	96.00	94.77	95.38	91.46

parameters discrepancy between the foundation model and the CD model, making it difficult for the CD model to absorb task-relevant knowledge from the larger model effectively. For the same model size, CLIP and RemoteCLIP demonstrate superior performance. The contrastive learning paradigm facilitates the transfer of comprehensive feature knowledge to downstream tasks, while RemoteCLIP more easily transfers domain-specific knowledge to downstream tasks in the remote sensing domain.

VI. CONCLUSION

In this article, we propose a novel burden-free KD method for efficient remote sensing CD BFD. It is a foundation model-based CD distillation framework designed to transfer general feature knowledge from foundation models to CD models. Compared to other foundation model-based CD approaches, BFD significantly enhances the performance of CD models without additional inference burden. Moreover, benefiting from the strong generalization ability of foundation models, BFD can achieve highly accurate results with limited training data. Trained with 40% labeled data, BFD surpasses the current SOTA semi-supervised CD methods. BFD is not limited to a specific foundation or CD model and exhibits excellent generalizability. However, BFD only utilizes the visual encoder of CLIP, leaving the richer information in the text encoder untapped. In the future, we will continue to explore how to efficiently adapt multimodal vision–language models to the remote sensing domain, leveraging textual descriptions for enhanced semantic understanding.

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